

ARGOS Weekly Update

TETHER: Causal Temporal Refinement for Frozen Surgical Stereo

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The Setting

The question

Can *one* causal temporal module be trained once and attached, unchanged, to heterogeneous **frozen** stereo estimators?

Constraints we accept

- No backbone internals.
- No future frames.
- Backbone weights never touched.

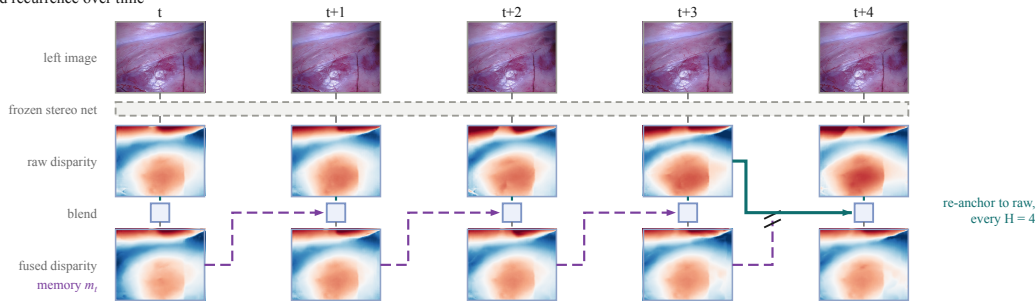
The module

154,874 parameters.

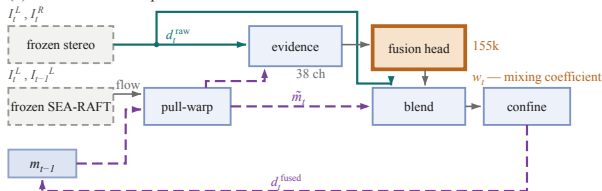
One scalar per pixel: how much to trust the past.

Architecture Overview

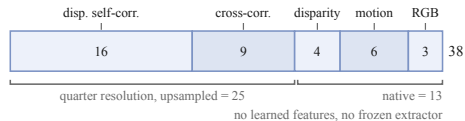
(a) bounded recurrence over time



(b) inside one time step



(c) evidence: what the 38 channels are



Grey is frozen. **One orange block is trained.** Memory is re-anchored to raw geometry every $H=4$ frames. $3/5$

It Transfers

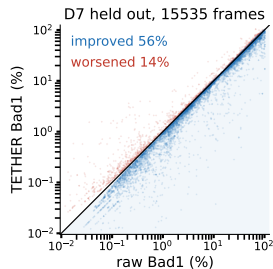
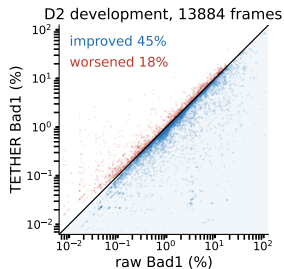
Mean EPE reduction

Backbone →	seen	unseen
SCARED-C D7	5.80%	4.79%
DRENDS (OOD)	5.57%	5.16%

One frozen checkpoint everywhere. Bottom-right is the cell usually left unevaluated.

Against a published stabiliser

Method	Causal	EPE	Bad1
Raw	—	.3029	2.09%
BiDASstabilizer	no	.3021	1.99%
TETHER	yes	.2889	1.80%



Every scored frame, all five backbones, no selection.
Below the diagonal is better.

Status

Paper

Eight pages, one figure, three tables, everything else in a seven-page supplement. Head-to-head comparison, three-seed study and the held-out transfer matrix are in.

The ablations came back, and not the way we expected

The learned feature extractor is not what makes the module work — removing it entirely is *better* and eight times more stable across seeds, which is why the shipped head is the smaller one. Relaxing the convex constraint then beat the canonical model on the held-out endpoint, which under our pre-registration means the claim that convexity explains cross-backbone transfer has to be withdrawn. It is withdrawn. We do not know why the head transfers.

Running

A 100-epoch convergence probe against the locked 12, to answer whether the budget is a ceiling or a plateau. Not eligible for selection.